项目名称：基于权限控制的校园数字证书链上存证系统——测试记录

测试环境：Ubuntu 20.04，FISCO BCOS 2.11.0，4 节点本地链。

主合约地址：0xe56915632c7ec33b092e72b4749f334da3bef6f8

测试用例 1：超级管理员发证

操作：教务处账户调用 createEvidence 传入合法证书哈希。

结果：status: 0x0，交易成功，数据成功上链。

```
[group:1]> call RbacCertificateEvidence 0xe56915632c7ec33b092e72b4749f334da3bef6f8 createEvidence "0xaaaa1111
1111111111111111111111111111111111111111111111111111111111"
transaction hash: 0x190d41927a7a8a63bd7ad2c9a3fa9e5ad3a2bca274b056197f42066e2b0a4b36
-------------------------------------------------------------------------------------------
transaction status: 0x0
description: transaction executed successfully
-------------------------------------------------------------------------------------------
Transaction inputs:
Input value size:1
Input types: (STRING)
Input values:(0xaaaa11111111111111111111111111111111111111111111111111111111111111)
-------------------------------------------------------------------------------------------
Receipt message: Success
Return message: Success
Return values:[]
-------------------------------------------------------------------------------------------
Event logs
Event: {}
```

测试用例：越权拦截重发

操作：未授权的黑客账户强行调用 createEvidence 伪造证书。

结果：触发 onlyAuthorized 拦截，报错 Not authorized to issue（status: 0x16），成功防御恶意篡改。

```
[group:1]> call RbacCertificateEvidence 0xe56915632c7ec33b092e72b4749f334da3bef6f8 createEvidence "0xaaaa1111
1111111111111111111111111111111111111111111111111111111111"
transaction hash: 0x190d41927a7a8a63bd7ad2c9a3fa9e5ad3a2bca274b056197f42066e2b0a4b36
-------------------------------------------------------------------------------------------
transaction status: 0x0
description: transaction executed successfully
-------------------------------------------------------------------------------------------
Transaction inputs:
Input value size:1
Input types: (STRING)
Input values:(0xaaaa11111111111111111111111111111111111111111111111111111111111111)
-------------------------------------------------------------------------------------------
Receipt message: Success
Return message: Success
Return values:[]
-------------------------------------------------------------------------------------------
Event logs
Event: {}

[group:1]> newAccount
AccountPath: account/ecdsa/0x86a3a94cb34e22de45e6cc3035a12744fc2d6c9f.pem
Note: This operation does not create an account in the blockchain, but only creates a local account, and depl
oying a contract through this account will create an account in the blockchain
newAccount: 0x86a3a94cb34e22de45e6cc3035a12744fc2d6c9f
AccountType: ecdsa

[group:1]> loadAccount account/ecdsa/0x86a3a94cb34e22de45e6cc3035a12744fc2d6c9f.pem
Load account account/ecdsa/0x86a3a94cb34e22de45e6cc3035a12744fc2d6c9f.pem success!

[group:1]> call RbacCertificateEvidence 0xe56915632c7ec33b092e72b4749f334da3bef6f8 createEvidence "0xbbbb1111
1111111111111111111111111111111111111111111111111111111111"
transaction hash: 0x68a6c3177bfae972c578dde9cd7e30e3ace3600f772c6a3f76a47a543c6c3b2c
-------------------------------------------------------------------------------------------
transaction status: 0x16
-------------------------------------------------------------------------------------------
Transaction inputs:
Input value size:1
Input types: (STRING)
Input values:(0xbbbb11111111111111111111111111111111111111111111111111111111111111)
-------------------------------------------------------------------------------------------
Receipt message: Not authorized to issue
Return message: Not authorized to issue
Return values:null
-------------------------------------------------------------------------------------------
```

测试用例：权限校验

操作：教务处账户调用 authorizeIssuer，授予黑客账户发证权限。

结果：status: 0x0，权限下发成功。

```
[group:1]> call RbacCertificateEvidence 0xe56915632c7ec33b092e72b4749f334da3bef6f8 authorizeIssuer 0x86a3a94c
b34e22de45e6cc3035a12744fc2d6c9f
transaction hash: 0x04bb56ec0e090842dee38b59224970933e6bd3160a286f57e825937c09201765
---------------------------------------------------------------------------------------------
transaction status: 0x0
description: transaction executed successfully
---------------------------------------------------------------------------------------------
Transaction inputs:
Input value size:1
Input types: (ADDRESS)
Input values:(0x86a3a94cb34e22de45e6cc3035a12744fc2d6c9f)
---------------------------------------------------------------------------------------------
Receipt message: Success
Return message: Success
Return values:[]
---------------------------------------------------------------------------------------------
Event logs
Event: {}
```

测试用例：授权下发

操作：获得授权的黑客账户再次调用 createEvidence。

结果：status: 0x0，成功将数据上链，证明权限控制流转有效。

```
[group:1]> call RbacCertificateEvidence 0xe56915632c7ec33b092e72b4749f334da3bef6f8 createEvidence "0xbbbb1111
1111111111111111111111111111111111111111111111111111111111111111"
transaction hash: 0x5514443164d40d5f4a1c767a2b1fc8b3d0eabdb6ce774995683c9d61479e2aad
---------------------------------------------------------------------------------------------
transaction status: 0x0
description: transaction executed successfully
---------------------------------------------------------------------------------------------
Transaction inputs:
Input value size:1
Input types: (STRING)
```

测试用例：5、公开防伪验证

操作：任何人调用 verifyEvidence 查询。

结果：返回 true 及发证地址和时间戳。

```
[group:1]> call RbacCertificateEvidence 0xe56915632c7ec33b092e72b4749f334da3bef6f8 verifyEvidence "0xbbbb1111
1111111111111111111111111111111111111111111111111111111111111111"
---------------------------------------------------------------------------------------------
Return code: 0
description: transaction executed successfully
Return message: Success
---------------------------------------------------------------------------------------------
Return value size:3
Return types: (ADDRESS, UINT, BOOL)
Return values:(0x86a3a94cb34e22de45e6cc3035a12744fc2d6c9f, 1794144055549, true)
---------------------------------------------------------------------------------------------
```

结论：系统成功实现了“未授权拦截-授权下发-授权发证-公开验证”的完整业务闭环，RBAC 机制运转无误。